

Why the outlet sponge fails to absorb the wake in the flow-past-cape ensemble, and why the ramp sweeps could not fix it

QG-Closure SGS track (diagnostic note)

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1 The reflection mechanism

The QG solver is pseudo-spectral, hence **periodic** in x . A flow-past configuration is emulated on this periodic torus by (i) a Brinkman volume penalty for the obstacle, (ii) a prescribed inflow speed $U(t)$, and (iii) an *outlet sponge* near the right boundary that is meant to absorb the outgoing wake. If the sponge fails to absorb it, the wake vorticity reaches the periodic boundary and **wraps around**, re-entering at the inflow as spurious upstream vorticity — the “reflection” seen in the truth Π fields. Because the solver is periodic, absorption at the outlet is the *only* thing preventing the wrap.

2 The sponge model and its absorption

The outlet sponge acts as a Rayleigh relaxation of the vorticity toward a datum (ω_{datum} , “return-to-datum”):

$$\partial_t \omega = \underbrace{-U(t) \partial_x \omega}_{\text{advection out}} - \underbrace{\eta \chi(x) (\omega - \omega_{\text{datum}})}_{\text{sponge}} + \dots \quad (1)$$

where $\chi(x) \in [0, 1]$ is the ramp mask (zero in the interior, rising to 1 at the outlet) and η the relaxation rate.

Step 0: the datum. For the *outlet vorticity* sponge the target is the clean far field, which carries no vorticity, so $\omega_{\text{datum}} = 0$. Then $\omega - \omega_{\text{datum}} = \omega$ and it drops from (1). Even for a nonzero (uniform) datum the result is unchanged: work with the *anomaly* $\omega' \equiv \omega - \omega_{\text{datum}}$; since the datum is a fixed target, $\partial_t \omega' = \partial_t \omega$ and $\partial_x \omega' = \partial_x \omega$, so (1) becomes $\partial_t \omega' = -U \partial_x \omega' - \eta \chi \omega'$. The “amplitude” below is this anomaly — the wake pulse we must absorb. We write ω for it hereafter.

Step 1: reduce to an ODE along a characteristic. With $\omega_{\text{datum}} = 0$, (1) is

$$\partial_t \omega + U \partial_x \omega = -\eta \chi(x) \omega. \quad (2)$$

The left-hand side is the material derivative along a fluid path $x(t)$ with $dx/dt = U$:

$$\left. \frac{d\omega}{dt} \right|_{\dot{x}=U} = \partial_t \omega + \dot{x} \partial_x \omega = \partial_t \omega + U \partial_x \omega = -\eta \chi(x(t)) \omega. \quad (3)$$

Step 2: integrate the ODE. Equation (3) is linear in ω , so

$$\omega(t) = \omega_0 \exp\left(-\int_0^t \eta \chi(x(t')) dt'\right). \quad (4)$$

Step 3: change the variable from time to position. Along the characteristic $dx' = U dt'$, i.e. $dt' = dx'/U$ (with U constant over one transit, see assumptions). Hence

$$\int_0^t \eta \chi dt' = \int_{x_0}^x \eta \chi(x') \frac{dx'}{U} = \frac{1}{U} \int_{x_0}^x \eta \chi(x') dx', \quad (5)$$

and (4) becomes the spatial amplitude profile

$$\omega(x) = \omega_0 \exp\left(-\frac{1}{U} \int_{x_0}^x \eta \chi(x') dx'\right). \quad (6)$$

Step 4: transmission and optical depth. The *transmission* T is the amplitude out over amplitude in across the whole sponge (entrance x_{in} , exit x_{out}). Evaluating (6),

$$T \equiv \frac{\omega(x_{\text{out}})}{\omega_0} = \exp\left(-\frac{1}{U} \int_{x_{\text{in}}}^{x_{\text{out}}} \eta \chi dx'\right) = e^{-\tau}, \quad \boxed{\tau(t) = \frac{1}{U(t)} \int_{\text{sponge}} \eta \chi(x) dx} \quad (7)$$

The optical depth τ is simply the exponent accumulated across the full sponge. A useful absorption target is $\tau \gtrsim 3$ (transmission $e^{-3} \approx 5\%$); at $\tau = 0.45$, $T = e^{-0.45} \approx 64\%$ survives. Two features of (7) are decisive: τ depends on the sponge only through the *integral* $\int \eta \chi dx$, and it is **inversely proportional to** $U(t)$. Since U is time-modulated, τ is smallest — and the leak largest — at the *high- U* phases.

Assumptions (this is a WKB / geometric-optics estimate). (i) U is taken constant during a single transit of the sponge; this is valid because the transit time $W/U \approx 2.37/2 \approx 1.2$ is short compared with the modulation period, so $\tau(t)$ is a quasi-static snapshot. (ii) In the sponge the wake is treated as a passively advected–damped scalar — the $-U\partial_x$ transport dominates, and dispersion and the elliptic ψ coupling are neglected there. (iii) The reduction is one-dimensional along the outflow-normal x . These are the standard sponge-layer assumptions; they are what collapse the absorption to the single number τ rather than a full 2-D calculation.

3 Measured quantities

(a) The inflow speed $U(t)$ is modulated, peaking at 2.87. From `U_snap` in each member (not an assumed constant):

member	U_{min}	U_{max}	U_{mean}
const	2.00	2.00	2.00
ou	1.88	2.46	2.16
ramp	2.00	2.87	2.60
sine	1.13	2.87	2.00
tel	1.13	2.87	2.28

(b) The outlet ramp (the actual reflection absorber). Measured from the FR mask `chi_sponge_ramp` along a mid-height row (so the separate top-wall sponge does not contaminate the x -integral):

property	value
location	$x \in [22.74, 25.11]$ u (last $W = 2.37$ u $\approx 0.1 L_x$)
shape	clean linear ramp $0 \rightarrow \chi_{\max} = 1.0$
rate	$\eta = 1.025$
integral	$\int \eta \chi_{\text{outlet}} dx = 1.28$

Check (triangle rule). For a linear ramp $0 \rightarrow 1$ over width W , $\int \chi dx = \frac{1}{2} \chi_{\max} W$, so

$$\int \eta \chi dx = \eta \frac{1}{2} \chi_{\max} W = 1.025 \times \frac{1}{2} \times 1.0 \times 2.37 = 1.21, \quad (8)$$

matching the numerically summed 1.28 (the small excess is the $\chi = 1$ plateau at the very edge). *The earlier $\tau \approx 1.9$ was wrong: it came from x -averaging over all y , which folds the top-wall sponge into the outlet integral. The honest outlet-only integral is 1.28.*

(c) The resulting optical depth is far below threshold.

$$\tau_{\text{outlet}} = \frac{1.28}{U} : \quad \tau(U=2.00) = 0.64, \quad \tau(U=2.87) = \mathbf{0.45}. \quad (9)$$

The sponge delivers **less than one e-folding** — transmission $T = e^{-0.45} \approx 64\%$ at the U peaks. Most of the wake passes straight through and wraps.

(d) Direct evidence: the reflection tracks $U(t)$. Correlation of the inflow vorticity $\|\omega\|_{\text{inflow}}(t)$ (the wrapped-back signal) with $U(t)$:

member	corr($\ \omega\ _{\text{inflow}}, U$)
ou	+0.40
ramp	+0.70
sine	+0.65
tel	+0.84

The correlation *rises with the modulation amplitude* (telegraph forcing, the largest swing, gives +0.84), exactly as $\tau \propto 1/U$ predicts: each high- U excursion drops τ and lets a pulse through.

4 Why the ramp sweeps could not fix it

To reach $\tau \geq 3$ at the worst case $U_{\max} = 2.87$ requires

$$\int \eta \chi dx \geq 3 U_{\max} = 3 \times 2.87 = 8.6, \quad \text{i.e. } \frac{8.6}{1.28} \approx \mathbf{6.7} \times \text{the present value.} \quad (10)$$

No single Rayleigh lever supplies $6.7\times$ within feasible bounds:

lever	required change	why it fails
width W	$2.37 \text{ u} \rightarrow 15.9 \text{ u}$	$= 0.63 L_x$; consumes the flow region
$\chi_{\max}$	—	already at 1.0; nothing left
rate η	$1.025 \rightarrow 6.9$	stiff+narrow sponge reflects off its own front

The last row is the subtle one: a Rayleigh layer that is too abrupt has an impedance mismatch at its leading face, so raising η trades the wrap-reflection for a sponge-face reflection. Widening enough to avoid that (the $W \rightarrow 16$ u route) eats the domain. **Hence sweeping ramp/width/ η was structurally unable to succeed:** with $\chi_{\max}$ already saturated, the only free parameter that could reach $\tau \geq 3$ is a width comparable to the whole domain.

Empirical confirmation: a width sweep (flow-past-cylinder). Sweeping the sponge width directly (leak advisory bar $\text{st_leak} < 0.05$; $v_{\text{rms}}, \omega_{\text{rms}}$ residual outlet fields) reproduces exactly this ceiling: every width fails the leak bar, performance is non-monotonic with an optimum near $W = 0.2$, it *reverses* beyond it, and the solver goes **unstable (NaN)** at $W = 0.3$ — long before any width capable of $\tau \geq 3$.

width W	v_{rms}	ω_{rms}	st_leak	verdict
0.025	.078	.0140	.66	fail
0.0375	.053	.0106	.82	fail
0.05	.040	.0134	.54	fail
0.075	.032	.0143	.76	fail
0.10	.045	.0132	.95	fail
0.125	.045	.0158	.73	fail
0.15	.028	.0135	.94	fail
0.20	.018	.0154	.85	fail (optimum)
0.25	.052	.0400	.85	fail (reversed 3 $\times$)
0.30	UNSTABLE — solver NaN at 7%, guard killed it in 7 min			

The optimum ($W = 0.2$) still leaks 85%, and the numerical wall (NaN at $W = 0.3$) arrives well before the width ~ 0.67 needed for $\tau \geq 3$. The width lever is capped by *instability*, not by absorption — exactly the τ ceiling, now observed directly.

5 Conclusion and the constraint on the fix

A Rayleigh outlet sponge *cannot* reach $\tau \geq 3$ in this geometry at these speeds: the domain is too short downstream and U too fast, and the width lever hits a numerical wall before it hits the absorption threshold.

The tempting fix is unavailable here. The natural non-reflecting outflow is a convective (radiation) condition,

$$\partial_t \omega + U(t) \partial_x \omega = 0 \quad \text{on an outlet strip,} \quad (11)$$

which lets vorticity *leave* at the local phase speed. It is non-reflecting by construction and needs no $\tau \geq 3$. **But (11) is a *local* boundary condition, and the solver is pseudo-spectral — globally periodic in x with no boundary to impose it on.** A hard reset of the field on the outlet strip is worse, not better: it is the $\eta \rightarrow \infty$ limit, so its infinitely sharp leading edge radiates a backward wave and injects Gibbs ringing through the FFT, and $v = \partial_x \nabla^{-2} \omega$ is nonlocal so it cannot be zeroed by fiat.

What the spectral framework actually allows. On a periodic torus the only tools are (i) volumetric relaxation terms (sponge / fringe / friction) and (ii) the domain size — and these are *exactly* what the τ ceiling bounds. The realistic remedies are therefore:

1. **Lengthen the downstream fetch.** A longer box lets the wake dissipate viscously before the wrap and gives a *wide, gentle* fringe room to reach $\tau \geq 3$ *without* eating the near-obstacle region or becoming numerically stiff. This is the spectral-compatible version of “absorb the wake.”
2. **A smooth fringe toward the inflow profile** ($\omega \rightarrow \omega_{\text{inflow}}$) over that longer fetch — still a volumetric term, hence spectral-legal, but graded enough to avoid the impedance mismatch that kills a stiff narrow sponge.
3. Failing a domain change, **accept the reflection** and handle it downstream in the closure (it is deterministic in the resolved state).

A genuine convective/radiation outflow would require a *non-periodic* treatment in x (e.g. a Chebyshev or finite-difference boundary layer) — a solver-level change, not a config flip.

Summary of the checks. (1) triangle rule $\int \eta \chi \, dx = \frac{1}{2} \eta \chi_{\text{max}} W = 1.21 \approx 1.28$ (measured); (2) mask floor is 0 in the interior, so there is *no* global-friction baseline — the outlet ramp is the whole absorber; (3) $\tau = \int \eta \chi \, dx / U$ gives $0.45\text{--}0.64 < 1$; (4) $\text{corr}(\text{reflection}, U)$ grows $+0.40 \rightarrow +0.84$ with modulation, confirming $\tau \propto 1/U$; (5) the $6.7\times$ deficit is unreachable by any single Rayleigh lever.